

Hypothesis-Search Report — ADNP-Related Syndrome

Hypothesis: NMD-Escaping Truncation Branch

(`nmd_escape_truncation_branch` , KB status: **ALTERNATIVE**)

OpenScientist autonomous run, 5 iterations. Search date: 2026-09-07. Primary tools: PubMed (search_pubmed), NCBI ClinVar E-utilities, Ensembl REST, gnomAD GraphQL. No patient-level data files were provided.

1. Executive Judgment

Verdict: PARTIALLY SUPPORTED / UNRESOLVED at the mechanistic-consequence level.

The hypothesis has two separable claims:

- **Claim A (upstream): "Most pathogenic last-exon truncating variants escape NMD and produce mutant transcript."** → **Established / strongly supported.** An independent ClinVar analysis performed here shows 150/165 (90.9%) pathogenic/likely-pathogenic truncating variants map to the last coding exon (or within 55 nt of the last exon–exon junction) and are predicted to escape NMD. Mutant *ADNP* mRNA has been directly detected in _____ patient _____ blood (_____).

[P 38926592](#)

The last coding exon carries ~94% of the CDS, so escape is the structural default for essentially any PTC past codon ~68.

- **Claim B (downstream): "Whether the effect is functional insufficiency, dominant interference, toxic gain of function, or a mixture remains unresolved because mutant protein has not been unambiguously demonstrated in patient material."** → **Unresolved — accurately stated.** Mutant *ADNP* protein has never been unambiguously demonstrated in _____ patient _____ tissue

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P 38926592
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P 36945042
 The ambiguity is a durable, decade-old open question (first raised in
P 25169753
 2014). Current model/in-vitro evidence *leans toward functional insufficiency* (last-exon knock-in mouse shows reduced Adnp levels and chromatin binding,
P 42208149
 nonsense variants reduce enzymatic activity,
P 41174994
 extreme LoF constraint pLI $\approx$ 1.0; a splice variant causes disease by pure haploinsufficiency,
P 38424297
),
 but a distinct/gain-of-dysfunction contribution is not excluded (overexpressed truncated mutants are expressed and mislocalize with mutation-specific phenotypes,
P 36230962
 /
 37759476; a missense variant acts dominant-negatively,
P 41943166
 ;
 another missense is predicted gain-of-function,
P 40977432
).

Bottom line: The hypothesis is a faithful, well-posed description of a genuinely unresolved mechanism. Its *premise* (NMD escape $\rightarrow$ mutant transcript) is confirmed; its *central question* (what the retained mutant protein does, if it exists at all in patients) remains open. Retaining ALTERNATIVE/unresolved status is appropriate. The most important caveat is that the balance of current evidence modestly favors haploinsufficiency/insufficiency as the dominant driver, with any gain-of-function/dominant-negative effect unproven at the endogenous patient-protein level.

2. Evidence Matrix

#	Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Summary
1	P 38926592 (D'Incal 2024)	Human/in vitro	Supports (A) / Qualifies (B)	NMD escape → mutant transcript → mutant protein?	Mutant <i>ADNP</i> mRNA detected in patient blood; validated ~150 kDa WT band; truncated mutant protein detectable only in <i>E. coli</i> /tagged overexpression, not in patient material	Las tru HV
2	This report — ClinVar computational analysis	Computational	Supports (A)	Do truncating variants cluster in last exon and escape NMD?	165/303 P/LP variants truncating; 150/165 (90.9%) predicted NMD-escape ; 15 NMD-sensitive are all codons 3–42	All var (GI
3	gnomAD v2.1.1 (this report)	Computational	Supports insufficiency (B)	Is <i>ADNP</i> dosage-sensitive/haploinsufficient?	pLI≈1.0; obs/exp LoF=4/84.9; LOEUF=0.11; LoF Z=7.45	Pop cor
4	P 42208149 (D'Incal 2026)	Model organism	Qualifies → insufficiency (B)	Effect of a last-exon frameshift knock-in in vivo	Reduced brain <i>Adnp</i> (p=8e-4), reduced chromatin association (p=1e-3), ↑ chromatin accessibility, Wnt	Ad p.L mo

#	Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Summary
					disruption, ASD-like behaviour	
5	P 41174994 (D'Incal 2025)	In vitro	Qualifies → insufficiency (B)	Do nonsense variants alter ADNP enzymatic activity?	ADNP methyltransferase activity reduced by nonsense variants; male-specific H3K79me1 reduction	Over + ti
6	P 36230962 (Ganaiem 2022)	In vitro	Supports gain-of-dysfunction (B)	Do truncated ADNP proteins behave as simple nulls?	GFP-tagged p.Ser404/p.Tyr719 expressed, accumulate in cytoplasm, mutation-specific phenotypes; aberrant nuclear/cytoplasmic boundaries	Tru mu ne
7	P 37759476 (Ganaiem 2023)	In vitro	Supports gain-of-dysfunction (B)	Mislocalization & microtubule effects of mutant ADNP	Aberrant nuclear/cytoplasmic ADNP distribution; reduced microtubule content; NAP rescues	Tru mu
8	P 38424297 (D'Incal 2024)	Human/in vitro	Competing → pure haploinsufficiency	Can HVDAS arise from LoF alone (no mutant protein)?	Splice-acceptor variant → exon-4 skipping; N-terminal truncated protein NOT detectable; "first confirmed diagnosis	No spl

#	Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Sub-claim
					exclusively due to haploinsufficiency"	
9	P 41943166 (Chen 2026)	In vitro + iPSC + mouse IUE	Competing → dominant-negative (missense)	Does mutant ADNP interfere with WT?	Missense p.C687R mislocalizes and redistributes wild-type ADNP ; disrupts GABAergic differentiation	Mislocalization of wild-type ADNP; HVDAS (HVDAS)
10	P 40977432 (Benvenuto 2026)	Human/computational	Competing → gain-of-function (missense)	Can ADNP variants strengthen function?	Missense p.Ser802Phe predicted to increase ADNP–DNA affinity → GoF hypothesis	Mislocalization of wild-type ADNP; HVDAS (HVDAS)
11	P 32758449 (Breen 2020)	Human clinical	Qualifies	Do the two mutation-position epesignatures map to phenotype?	Two mutation-dependent epesignatures replicated; limited phenotype correlation ; no profound blood transcriptome change	HVDAS vs. HVDAS
12	P 38884529 (Sarli 2024)	Human clinical	Qualifies	Is "class II" a distinct truncating subtype?	Class II HVDAS (truncating, BNL-region) shares a phenotype-specific epesignature with SMARCA2/BIS	Class II HVDAS (HVDAS)
13	P 38637827 (D'Incal 2024)	Human tissue	Supports downstream effect	Downstream consequences of a last-exon	Cerebellar CpG methylation changes + mitochondrial	HVDAS (HVDAS)

#	Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Summary
				frameshift in patient brain	gene dysregulation; ADNP motif enriched in DMGs	
14	P 25169753 (Vandeweyer 2014)	Human clinical	Supports framing (B)	Origin of GoF vs haploinsufficiency debate	First 10 truncating mutations "suggested a gain of function"; 11th "predicted haploinsufficient"	First column
15	P 29724491 (Van Dijck 2019)	Human clinical	Context	Clinical phenotype of HVDAS	78-patient deep clinical characterization; multisystem syndrome	HVDAS
16	P 36945042 (D'Incal 2023)	Review	Supports (seed)	Defines the unresolved NMD-escape branch	States mutant protein never unambiguously demonstrated; toxic GoF "could be envisaged"	Review

3. Data and Tool Use

Databases / APIs accessed

Source	Accession/URI	Version/ snapshot	Retrieval date	Outcome	Query / filters
NCBI ClinVar (E-utilities)	https://eutils.ncbi.nlm.nih.gov/entrez/eutils/ (db=clinvar)	live DB (no pin)	2026-09-07	succeeded	ADNP[gene] AND ("pathogenic"[clinical research] OR "likely pathogenic"[clinical research])→303 IDs; eGRCh38 positions
Ensembl REST	https://rest.ensembl.org/lookup/id/ENST00000621696?expand=1	Ensembl (2026-09)	2026-09-07	succeeded	ADNP canonical MA transcript exon/CDS coordinates
gnomAD GraphQL	https://gnomad.broadinstitute.org/api	gnomAD v2.1.1 constraint	2026-09-07	succeeded	gene(gene_symbol) constraint fields
ClinGen	https://search.clinicalgenome.org/kb/genes/HGNC:15766;/api/curations/HGNC:15766;/api/dosage/HGNC:15766	n/a	2026-09-07	partial/failed	Gene page HTTP 200 SPA, "ADNP curation" JSON API endpoints 404
PubMed (search_pubmed)	—	—	2026-09-07	succeeded	12 short queries (low bundle); 4 over-specified queries returned 0

Analyses (input → method → output)

Analysis 1 — ADNP variant NMD-escape mapping. Outcome: SUCCEEDED. - Input: 303 ClinVar P/LP ADNP variants (GRCh38 positions) + ADNP exon/CDS structure (ENST00000621696, chr20 minus strand; last coding exon 50,888,918–50,894,512). - Method: classify variants by ClinVar molecular_consequence/title; map genomic position to exon; predict NMD escape if PTC in last coding exon or within 55 nt upstream of last exon–exon junction. - Environment: python 3.12; pandas 3.0.3; numpy 2.3.5; requests 2.34.2; matplotlib 3.10.9. No random seed (deterministic given DB snapshot). - Output: `openscientist_artifacts/tables/summary_stats.json`, `class_counts.csv`, `nmd_sensitive_early_variants.csv`; code in `code/adnp_clinvar_nmd_analysis.py`. Figure generated in code sandbox but **executor-local only** (separate container; not persisted to job FS) — regenerate via the code. - Result: 165 truncating; 150 (90.9%) predicted NMD-escape. - Limitations: NMD escape predicted, not measured; ClinVar submission bias; class inferred from text fields.

Analysis 2 — gnomAD constraint lookup. Outcome: SUCCEEDED. - Input → Method → Output: gnomAD GraphQL → parse constraint → `tables/gnomad_adnp_constraint.json`. pLI=1.0, LOEUF=0.11. - Limitation: LOFTEE pLoF mostly captures NMD-competent variants; does not directly quantify the last-exon patient class.

Analysis 3 — ClinGen dosage/validity retrieval. Outcome: FAILED (reported, not silently dropped). - API JSON endpoints 404; gene page is an HTML SPA. No fallback to model knowledge; classification left unverified. Logged in `logs/iter3_4_search_log.txt`.

Artifact bundle root: `kb/hypotheses/ADNP-Related_Syndrome/nmd_escape_truncation_branch/openscientist_artifacts/` (MANIFEST.yaml, code/, tables/, logs/, environment/). Input checksums not computed (hashlib blocked in sandbox); inputs are regenerable from the pinned queries.

4. Mechanistic Causal Chain

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De novo heterozygous ADNP variant
| [STRONG] ~55% truncating (frameshift/nonsense), ~35% missense (ClinVar, this report)
▼
Truncating PTC in last coding exon (≥94% of CDS)
| [STRONG] 90.9% predicted NMD escape (this report); mutant mRNA in patient blood (<a href="https://pubmed.ncbi.nlm.nih.gov/32751416/">https://pubmed.ncbi.nlm.nih.gov/32751416/
▼
Mutant transcript retained → [MISSING/UNPROVEN] Endogenous mutant protein in patient tissue
| (undetectable to date; <a href="https://pubmed.ncbi.nlm.nih.gov/32751416/">https://pubmed.ncbi.nlm.nih.gov/32751416/
▼
Net functional consequence = one (or a mix) of:
|─ Functional insufficiency [MODERATE support: <a href="https://pubmed.ncbi.nlm.nih.gov/42208111/">https://pubmed.ncbi.nlm.nih.gov/42208111/
|─ Dominant interference [INFERRED: <a href="https://pubmed.ncbi.nlm.nih.gov/41943166/">https://pubmed.ncbi.nlm.nih.gov/41943166/ re
|─ Toxic gain of function [SPECULATIVE: <a href="https://pubmed.ncbi.nlm.nih.gov/40977432/">https://pubmed.ncbi.nlm.nih.gov/40977432/
|─ Mixture [PLAUSIBLE, untested at endogenous level]
▼
Impaired ADNP chromatin regulation (ChAHP/SWI-SNF; CTCF antagonism) + cytoskeletal/microtubule c
| [STRONG in models] ↑chromatin accessibility, Wnt disruption (<a href="https://pubmed.ncbi.nlm.nih.gov/32751416/">https://pubmed.ncbi.nlm.nih.gov/32751416/
▼
Altered gene-expression & DNA-methylation programs (episignatures; mitochondrial genes)
| [STRONG] two mutation-position episignatures (<a href="https://pubmed.ncbi.nlm.nih.gov/32751416/">https://pubmed.ncbi.nlm.nih.gov/32751416/
▼
Aberrant neurodevelopment (neurogenesis, GABAergic differentiation, synaptic pruning)
▼
HVDAS clinical phenotype: ASD, ID, dysmorphism, multisystem features (<a href="https://pubmed.ncbi.nlm.nih.gov/32751416/">https://pubmed.ncbi.nlm.nih.gov/32751416/

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Strong links: variant→NMD escape→mutant transcript; ADNP loss→chromatin/expression changes→neurodevelopmental phenotype (models + patient tissue + episignature). **Inferred links:** mutant transcript→which protein-level consequence; dominant-negative/GoF edges rest on missense or overexpression data. **Missing step (the crux):** unambiguous demonstration (and quantification) of endogenous mutant protein in patient material, and whether it is null-equivalent, interfering, or toxic.

5. Knowledge Gaps

1. **Endogenous mutant protein — existence and behavior (CRITICAL, checked).** No study has unambiguously detected truncated ADNP in patient tissue

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P 38926592

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Resolves via: sensitive targeted mass spectrometry (PRM/MRM for mutant-specific tryptic/neo-C-terminal peptides) in patient iPSC-neurons/organoids or accessible tissue; allele-specific ribosome profiling.

2. **Insufficiency vs dominant-negative vs toxic gain (CRITICAL).** Models lean to insufficiency but overexpression/missense data keep interference/GoF alive. Resolves via: isogenic patient vs corrected iPSC-neurons at endogenous expression; degron/dose-titration; co-expression interference assays with tagged WT.

3. **Subtype heterogeneity by variant position.** Two epesignatures exist but correlate weakly with phenotype

(P 32758449);

different truncations give opposite in-vitro phenotypes

(P 36230962).

Whether "class I/II" reflect distinct mechanisms is unresolved. Resolves via: position-stratified functional cohorts.

4. **Constraint metric vs patient variant class.** gnomAD pLI $\approx$ 1.0 but LOFTEE mostly scores NMD-competent variants; the last-exon patient class is under-represented in the metric. A last-exon-specific constraint/patient-recurrence analysis is missing.
5. **ClinGen/GenCC source-level absence (checked, unverified).** ClinGen ADNP curation page resolves but dosage/validity JSON returned 404 on 2026-09-07; classification could not be programmatically confirmed. Label: **unverified**, not absent. Resolves via: manual ClinGen/GenCC lookup by a curator.
6. **Treatment mechanism (NAP/davunetide).** NAP rescues mutant-cell phenotypes

(P 37759476)

but its precise molecular target relative to the mutant protein is not established; no completed HVDAS-specific efficacy trial identified in this search.

6. Alternative Models

- **Haploinsufficiency / functional insufficiency (KB-likely PRIMARY).** *Alternative to the seed's GoF/interference branches; also the parsimonious default.* Supported by ↓ protein/activity in models, extreme LoF constraint, and a pure-HI splice case

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P 38424297

P 42208149

P 41174994

The seed hypothesis is compatible with this if the mutant protein is null-equivalent.

- **Dominant-negative interference.** *Alternative downstream branch of the seed.* Missense p.C687R redistributes WT ADNP

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P 41943166

truncated mutants mislocalize

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P 36230962

Endogenous evidence lacking.

- **Toxic gain of function.** *Alternative downstream branch.* In-silico ↑ DNA affinity

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P 40977432

mutation-specific cellular phenotypes. Most speculative.

- **Cytoskeletal/microtubule-autophagy dysfunction (parallel mechanism).** ADNP-EB1/EB3/Tau axis; mTOR/autophagy (everolimus rescue in KD mouse). Downstream/parallel to chromatin dysfunction rather than competing at the variant level.

- **Cell-type-specific mechanisms (parallel).** ADNP roles in microglial synaptic pruning

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and GABAergic differentiation

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P 41943166

suggest non-neuronal contributions.

7. Discriminating Tests

1. **Endogenous mutant-protein detection (decisive).** Targeted PRM/MRM mass spectrometry + allele-specific Western/immunoprecipitation on patient iPSC-derived neurons/organoids carrying recurrent last-exon variants (e.g., p.Tyr719, p.His559fs). Expected:* insufficiency →

little/no stable mutant protein; interference/GoF → detectable stable truncated protein.
Directly closes the crux gap.

2. **Isogenic dose-response.** Patient vs CRISPR-corrected isogenic iPSC-neurons; compare to heterozygous null (early-exon NMD-competent variant) at matched differentiation.
Expected: if phenotype of last-exon variant = heterozygous null → insufficiency; if more severe or qualitatively different → interference/GoF.
3. **WT-interference assay.** Co-express endogenous-level tagged WT with mutant; quantify WT chromatin occupancy (CUT&RUN) and localization. *Expected:* dominant-negative → reduced WT occupancy/mislocalized WT.
4. **Position-stratified multi-omic cohort.** Stratify patients by class I vs II episignature and by variant position; RNA-seq/methylation on iPSC-neurons. *Expected:* distinct molecular consequences would justify subtype restriction of the hypothesis.
5. **Allele-specific ribosome profiling** in patient cells to confirm translation of the mutant transcript (bridges transcript→protein step).

8. Curation Leads (require curator verification)

Candidate	evidence	references	+	snippets	to	verify:	-
P 38926592	— "we previously demonstrated escape from nonsense-mediated decay by detecting mutant ADNP mRNA in patient blood" (upstream NMD-escape). <i>Note:</i> the seed's snippet for						
P 36945042	is from full text, not the stored abstract — verify against full text.						
P 42208149	— "reduced cellular Adnp levels in the brain (p = 0.0008) and decreased its chromatin association (p = 0.001)" (insufficiency edge).						
P 25169753	— "the first 10 patients ... suggested a gain of function mechanism, an 11th patient ... is predicted haploinsufficient" (durable ambiguity).						
P 36230962							

— "these discrete phenotypes were associated with an increased expression of both mutant proteins in the cytoplasm" (gain-of-dysfunction lead; overexpression caveat). -

P 38424297

— "first confirmed diagnosis exclusively due to haploinsufficiency of the ADNP gene" (pure-HI competing model).

Candidate pathophysiology nodes/edges: add edge `mutant transcript → (unproven) endogenous mutant protein` with status *unconfirmed*; edges `truncating variant → NMD escape (predicted, 90.9%)`, `ADNP LoF → ↑chromatin accessibility/CTCF antagonism`, `ADNP LoF → episignature (class I/II)`, `ADNP LoF → mitochondrial gene dysregulation (patient cerebellum)`.

Candidate ontology terms: GABAergic interneuron differentiation (GO:0097154); microglial synaptic pruning (GO:0150062-related); microtubule cytoskeleton organization (GO:0000226); chromatin remodeling (GO:0006338); nonsense-mediated decay (GO:0000184). Cell types: cortical neural progenitor, GABAergic interneuron, microglia, cerebellar tissue.

Candidate subtype restriction: hypothesis applies specifically to last-exon truncating variants (~91% of truncating, ~50% of all P/LP alleles); explicitly does NOT cover early NMD-competent variants (codons 3–42; the pure-HI splice case) or missense variants (which have their own dominant-negative/GoF evidence).

Candidate status: retain ALTERNATIVE / unresolved. Add `knowledge_gaps`: (i) endogenous mutant protein undemonstrated; (ii) insufficiency vs interference vs GoF unresolved; (iii) ClinGen/GenCC classification unverified as of 2026-09-07 (API 404).

9. Limitations

- Literature-and-public-database study; no patient-level omics data were provided or analyzed.
- NMD escape is *predicted* (last-exon/55-nt rule), not experimentally measured per variant.
- ClinVar is a live, submission-biased database (no version pin); counts may drift.
- gnomAD constraint reflects NMD-competent pLoF, not the last-exon patient class.
- ClinGen/GenCC classifications could not be programmatically verified (endpoints 404).

- Several mechanistic in-vitro results rely on overexpressed/tagged constructs, not endogenous patient protein.
- Analysis figure is executor-local (separate sandbox container) and was not persisted to the bundle; it is regenerable from the included code.

Findings recorded to the knowledge graph this run: 7. Artifact bundle: `kb/hypotheses/ADNP-Related_Syndrome/nmd_escape_truncation_branch/openscientist_artifacts/`.

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